Pig Training — Trainer Guide

A step-by-step guide for running a pig training session. Written for day-to-day trainers, not programmers. You should not need to touch a command line or a code file.

If anything looks wrong, **stop and call the lab tech**. Don't try to debug terminal output yourself.

What the system does

The training rig combines two pieces of equipment:

```
+--------------------------+        +-----------------+
|    Touchscreen + PC      |  USB   |    MED-PC       |
|    (this software)       | <----> |    + ENV-204    |
|                          |        |    pellet feeder |
+--------------------------+        +-----------------+
       (you sit here)                 (in the same rack)
```

- The **touchscreen PC** runs the training task. It shows buttons, plays tones, records responses.
- The **MED-PC computer** drives the pellet feeder. For tasks that reward with a pellet, the touchscreen PC tells MED-PC when to drop one.

The three tasks

There are three trainer-runnable tasks in this system:

Task	Pellets?	Use it for
Shaping (full)	Yes — MED-PC pellet feeder	Teaching new subjects to touch the screen, all the way through to fine motor accuracy. Five stages from autoshape → punish-incorrect.
Motor Task — Accuracy	No (touchscreen only)	Fine motor accuracy testing on subjects already trained. Saves a CSV of touch coordinates and accuracy.
Motor Task — Reaction Time	No (touchscreen only)	Reaction-time testing. Saves a CSV of stimulus-to-touch latencies.

You pick which task in the launcher (next section). Only **Shaping** requires MED-PC + a stocked hopper.

> **Known issue with the two Motor Task scripts:** they may stop partway through a session on the current Python version. If you see the touchscreen freeze or the program close unexpectedly, **stop and call the lab tech** — don't restart the session blind. Shaping is not affected.

Before each session

For all tasks

- [ ] Touchscreen is on and you can see the desktop.
- [ ] Subject is at the rig and able to reach the screen.
- [ ] You have ~30–60 minutes of uninterrupted time.

For Shaping (pellet) sessions, also:

- [ ] **Hopper is full** — at least ~150 pellets in the ENV-204.
- Open the feeder lid and top up.
- [ ] **MED-PC.exe is running** on the rack computer.
- If the MED-PC screen is blank or asleep, wake the machine.
- [ ] **MSN is started** via MPCLoader.
- Open MPCLoader, run the loader macro (your lab will have an icon or a saved shortcut), then click **S;1**.
- The MED-PC display should show `Box 1` running with empty counters: `Verified-sess=0`, `Last-delivered=0`, `Empty-hopper-errs=0`.

If any of these aren't ready, **don't start the session** — fix them first or ask the lab tech.

Starting a session

Step 1 — Open the launcher

Double-click **Start Training** on the desktop.

Two things happen:

1. A black **terminal window** opens. Leave it alone — it shows the reward log while the session is running.
2. A small **Pig Training** window appears on top:

```
+-------------------------------------------------+
|   Pig Training                                  |
|   Pick a task and click Start.                  |
|                                                 |
|   Task:                                         |
|   [ Shaping (MED-PC pellet)      v ]            |
|                                                 |
|   Notes:                                        |
|   +-------------------------------------------+ |
|   | Uses the MED-PC pellet feeder.          |   |
|   |                                         |   |
|   | Before starting:                        |   |
|   |   - MED-PC.exe must be running          |   |
|   |   - In MPCLoader, run the loader macro  |   |
|   |     and click S;1                       |   |
|   |   - Hopper should hold at least 150     |   |
|   |     pellets                             |   |
|   +-------------------------------------------+ |
|                                                 |
|        [   Start Training   ]                   |
|                                                 |
+-------------------------------------------------+
```

Step 2 — Pick a task

Click the dropdown, choose the task you want. The notes box updates to show what each task needs.

Step 3 — Click "Start Training"

The launcher window closes. After a moment, a **settings window** appears (this is the task's own configuration screen).

Configuring the settings

Each task has its own settings popup. The fields are similar.

Shaping (full)

```
+------------------------------------------------------------------------+
|  Stage:                Responses:    Timers (seconds):                 |
|  ( ) Stage 0 (FR-1)    Stage 0: 20   Reinforcer Delay: 30              |
|  ( ) Stage 1           Stage 1: 40   Limited Hold:     25              |
|  ( ) Stage 2           Stage 2: 15   Blackout:         0.15            |
|  ( ) Stage 3           Stage 3: 40   FR Requirement:   1               |
|  ( ) Stage 4           Stage 4: 50                                     |
|                                                                        |
|  Autoshape: [ Yes v ]                                                  |
|                                                                        |
|                          [   Start   ]                                 |
+------------------------------------------------------------------------+
```

Field	What it does
Stage	Which stage the subject starts in. Stage 0 is autoshape (no responses required); Stage 4 punishes incorrect touches. Pick where the subject is in their training trajectory.
Responses (per stage)	How many correct touches before the task advances to the next stage. Defaults are calibrated for typical first-time subjects.
Reinforcer Delay	How long the screen waits before delivering the pellet on autoshape.
Limited Hold	How long the subject has to respond before the trial counts as an omission.
Blackout	How long the screen stays black between trials, in minutes.
FR Requirement	Fixed Ratio — how many presses count as one response.
Autoshape	If "Yes", Stage 0 automatically delivers a pellet after the Reinforcer Delay even without a press. Leave on "Yes" for new subjects.

Click **Start** when you're done. The screen goes fullscreen black for the Blackout duration, then the first trial begins.

Motor Task — Accuracy / Reaction Time

These tasks have a similar settings screen with response counts and timers. The defaults are usually fine. The Motor Tasks do not deliver pellets — they only score the subject's touches.

During the session

What to watch

- The **touchscreen** shows the trial buttons. The subject does the work here — don't touch the screen during a trial.
- The **terminal window** shows one line per delivered pellet, like:

`trial started - stage 0 [REWARD] trial 1: requested=1 delivered=1 status=ok`

Each `[REWARD]` line means one pellet was successfully delivered. The Motor Tasks don't print `[REWARD]` lines (no pellets).

What you should NOT do

- Don't touch the keyboard.

- Don't click anywhere on the touchscreen unless the task is in a stage that explicitly waits for a hand-shape (most of the time it doesn't).
- Don't close the terminal window.
- Don't open another program on the same desktop.

How long it takes

Stage	Approximate duration
Stage 0 (Shaping, autoshape)	~15 s per trial × 20 trials = ~5 min
Stage 1	~10 s per trial × 40 = ~7 min
Stage 2	~10 s per trial × 15 = ~3 min
Stage 3	~15 s per trial × 40 = ~10 min
Stage 4	~20 s per trial × 50 = ~17 min
Full Shaping session, all stages	~40–50 min

A Motor Task session is typically 15–30 min, depending on the response count.

Ending a session

When the session ends on its own

After the last trial:

1. A **report screen** appears showing per-stage statistics:

```
+----------------------------------------------------+ | Stage 0 Stage 1 Stage 2 Stage 3 Stage 4 | | Resp: 20 Resp: 40 Resp: 15 Resp: 40 Resp: 50 | | Inc: 12 | | Omit: 3 | | | | [ Exit ] | +----------------------------------------------------+
```

2. Click **Exit** to close the report.
3. The terminal window shows `Disconnected!` and waits.
4. Press any key to close the terminal.

Ending early

If you need to stop a session before it completes:

1. Find the **tiny gray square** in the top-left corner of the touchscreen (about the size of a pencil eraser tip). Click it.
2. The report screen appears with whatever stats accumulated so far.

3. Proceed as above.

Where the data is saved

Task	Data location
Shaping (Shaping_full.py)	`C:\MED-PC\Data\` on the MED-PC computer (auto-saved when the session ends; one file per session)
Motor Task — Accuracy	`HamTraining\Data\*.csv` on the touchscreen PC
Motor Task — Reaction Time	`HamTraining\Data\*.csv` on the touchscreen PC

Talk to the lab tech before moving or deleting these files.

Quick checklist (for daily use)

The reference card condenses this guide into a single printable page. Print it and tape it to the rig.

Glossary

- **MED-PC** — Med Associates Inc.'s control software for the pellet feeder hardware. Runs on its own computer in the rack.
- **MSN** — MED-PC State Notation. The "program" MED-PC runs. The one this lab uses is called *CoasterChase*.
- **MPCLoader** — The MED-PC utility used to load and start an MSN.
- **S;1** — The MPCLoader command that starts the MSN in Box 1.
- **ENV-204 / VeriFEED** — The pellet feeder hardware. Includes an IR sentry that confirms each pellet was actually delivered.
- **Hopper** — The pellet reservoir on the feeder. Holds a few hundred pellets when full.
- **Autoshape** — A training stage where the system delivers a pellet automatically on a schedule, even without a subject response. Used to bootstrap a new subject's screen-touching behavior.
- **FR (Fixed Ratio)** — The number of responses required per reinforcement. `FR-1` means one press per pellet.